

IntentTalk: Future-Grounded Motion Planning for Event-Aware Co-Speech Motion Generation

Jiye Zhang, Guibiao Liao, Dingwei Liu, Gaolin Yang, Xiuhua Jiang, and Jiangbo Xu

Abstract—Co-speech gesture generation has made substantial progress in motion quality and speech–motion synchronization, yet existing methods predominantly map speech directly to motion or enhance semantic information at individual temporal positions. Such formulations provide limited explicit modeling of how a semantically relevant gesture should evolve before and around its occurrence. We propose IntentTalk, a hierarchical speech-to-plan-to-motion framework that introduces a motion-grounded discrete plan as an intermediate representation between speech understanding and holistic motion generation. During training, a future motion-plan tokenizer encodes upcoming body-motion windows into discrete plan tokens, providing future-grounded supervision without requiring manually defined gesture categories or intention labels. A speech-to-plan predictor subsequently learns to infer these plans solely from speech-semantic features, enabling future-aware planning at inference without access to future motion. The predicted plans guide generation through two complementary mechanisms: an event-aware semantic gate refines *when* semantic body motion should be activated, while hierarchical plan-conditioned FiLM controls *how* the planned motion is realized across shared, body-part, and motion-token representations. Extensive experiments on BEAT2 and GES-X datasets demonstrate that IntentTalk achieves superior performance over representative state-of-the-art methods, validating the effectiveness of the proposed future-grounded motion planning framework.

Index Terms—Co-speech gesture generation, future motion planning, discrete motion representation.

I. INTRODUCTION

CO-SPEECH gestures complement verbal communication by conveying rhythmic, semantic, and expressive information, making them essential for realistic virtual humans and embodied agents. Recent Co-speech gesture generation methods have achieved substantial progress through discrete motion representations, diffusion-based generation, and efficient sequence modeling [1]–[5]. Meanwhile, semantic-aware approaches incorporate linguistic cues, semantic priors, retrieval mechanisms, or explicit relevance modeling to improve speech–gesture correspondence [6]–[10]. Despite these

advances, generating gestures that are simultaneously semantically appropriate, temporally organized, and structurally coherent remains challenging.

Most existing methods directly map speech representations to detailed motion, requiring a single generation process to jointly determine high-level gesture organization and low-level kinematic realization. Semantic-aware methods partially alleviate this difficulty by detecting relevant segments, retrieving gesture priors, or injecting higher-level semantic conditions [6], [7], [10], [11]. However, such semantic guidance primarily helps determine *when* speech content should influence motion, but does not explicitly model *how* an upcoming gesture should evolve over time. Since a meaningful gesture typically unfolds over multiple frames through preparation, salient movement, and transition, current-state semantic conditioning can improve the semantic relevance of motion around the current time step, while leaving the temporal organization of the upcoming gesture largely implicit, as illustrated in Fig. 1(a–b).

Recent work has demonstrated the benefit of incorporating future constraints. In particular, StreamTalk [12] introduces a closed-loop generate–retrieve–refine strategy, where a plausible tail key pose retrieved from a motion database serves as a forward anchor to reduce long-horizon motion drift. This suggests that forward-looking motion constraints can help stabilize subsequent motion evolution. Nevertheless, such a key-pose anchor primarily specifies a geometric destination, namely *where* the motion should arrive, rather than explicitly representing *how* the upcoming gesture should evolve over a temporal interval according to speech context. We therefore distinguish *future anchoring* from *future planning*: the former constrains a target motion state, whereas the latter aims to model the temporally extended organization of an upcoming gesture before its detailed realization.

The need for explicit future planning is further motivated by the inherent ambiguity of speech-to-gesture mapping. Similar speech content may correspond to multiple plausible motions involving different body regions, amplitudes, trajectories, and temporal patterns [2], [13], [14]. Directly resolving this ambiguity at the level of detailed motion places a substantial burden on the generator and may result in locally plausible motion with weak temporal organization. We hypothesize that an intermediate representation of upcoming motion organization can provide a more structured generation process. Instead of immediately predicting detailed joint motion, the model first anticipates a coarse, motion-grounded future plan from speech and then realizes it into fine-grained motion. This yields the hierarchical *speech-to-plan-to-motion* formulation illustrated in Fig. 1(c).

Based on this motivation, we propose IntentTalk, a hierar-

Jiye Zhang is with the School of Information and Communication Engineering, Communication University of China, Beijing 100024, China and Pengcheng Laboratory, Shenzhen 518108, China (e-mail: zhangjiye@cuc.edu.cn).

Guibiao Liao is with School of Computing and Data Science, The University of Hong Kong, Hong Kong SAR, China and Hong Kong Generative AI Research and Development Center, Hong Kong SAR, China (e-mail: gbliao@hku.hk).

Dingwei Liu is with the School of Future Technology, South China University of Technology, Guangzhou 511442, China and Pengcheng Laboratory, Shenzhen 518108, China (e-mail: ftliudw@mail.scut.edu.cn).

Xiuhua Jiang, Jiangbo Xu and Gaolin Yang are with the School of Information and Communication Engineering, Communication University of China, Beijing 100024, China (e-mail: jiangxiuhua@cuc.edu.cn; xujiangbo@cuc.edu.cn; yanggaolin@cuc.edu.cn).

Corresponding author: Jiangbo Xu

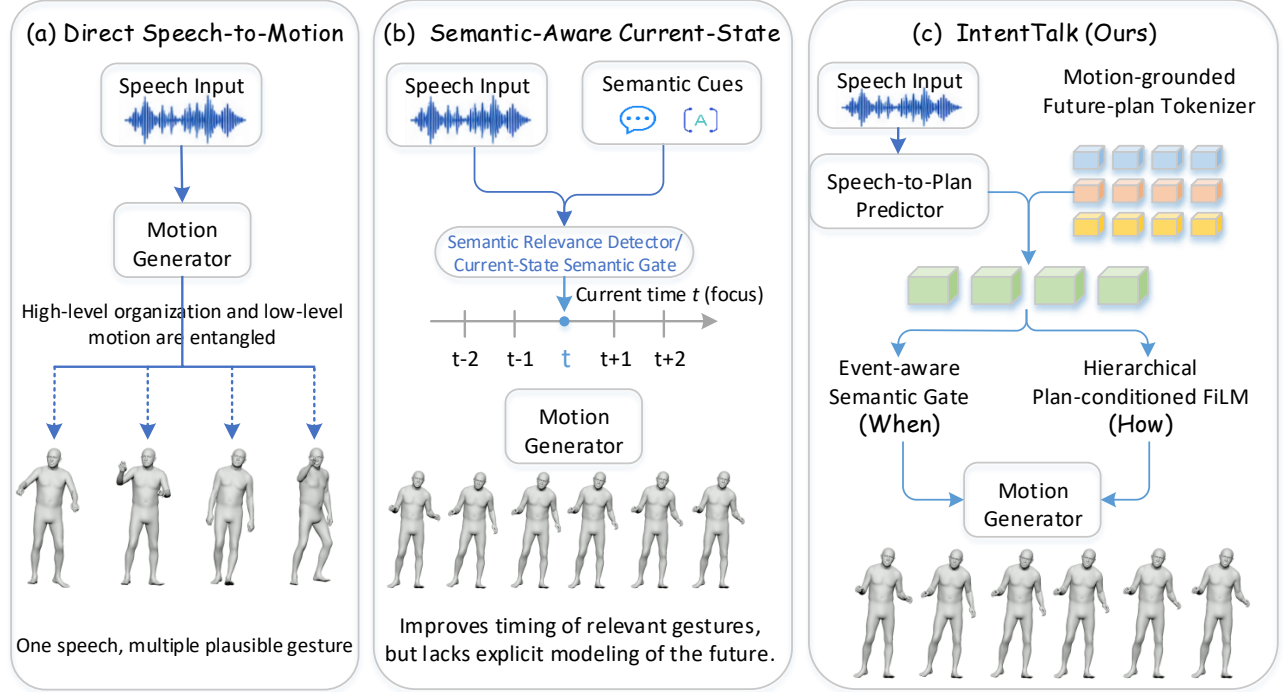


Fig. 1. Comparison of representative co-speech gesture generation paradigms. Direct speech-to-motion generation entangles high-level gesture organization with low-level realization, while semantic-aware methods mainly provide current-state relevance guidance. IntentTalk instead predicts a speech conditioned, motion-grounded future plan before detailed motion generation, explicitly separating future gesture organization from its kinematic realization.

chical co-speech gesture generation framework with explicit *speech-conditioned future motion planning*. During training, a motion-plan tokenizer encodes an offset window of upcoming body motion into a compact discrete plan representation, providing motion-grounded supervision for learning representative future motion structures. By summarizing the temporal organization of an upcoming motion segment rather than a single target state, the learned plan provides structured guidance for subsequent motion realization. A speech-to-plan predictor then learns to infer the corresponding plan from speech-semantic features. Importantly, future motion is used only to construct planning supervision; at inference, the plan is predicted entirely from observed speech, without requiring future motion or external motion retrieval. The predicted plan guides generation through two complementary mechanisms: an *event-aware semantic gate* determines *when* semantically relevant motion should be activated, while *hierarchical plan-conditioned FiLM* controls *how* the planned motion is realized across shared, body-part, and motion-token representations. Together, these mechanisms integrate the predicted future plan into both temporal activation and multi-level motion realization. In this way, IntentTalk explicitly separates future motion planning from low-level kinematic realization.

We evaluate IntentTalk on BEAT2 [1] and further examine cross-dataset generalization on GES-X [15]. Beyond standard measures of motion quality, diversity, synchronization, and semantic relevance, we analyze the learned planning representation through current-versus-future comparisons, event-level evaluation, planning-quality analysis, and controlled plan interventions. These experiments examine both generation quality and whether explicit future planning provides comple-

mentary guidance beyond current-state semantic conditioning and geometric future anchoring.

Our main contributions are threefold:

- We propose IntentTalk, a hierarchical *speech-to-plan-to-motion* framework that introduces explicit speech conditioned planning of upcoming gesture structure before detailed motion generation.
- We develop a motion-grounded discrete future-plan representation learned from upcoming motion windows, together with a speech-to-plan predictor that anticipates these latent plans from speech alone at inference without requiring future motion or retrieved destination poses.
- We introduce an event-aware and hierarchical plan-conditioned generation mechanism that exploits the predicted plan at complementary temporal and representational levels, regulating both *when* semantic motion should be activated and *how* the planned gesture should be realized.

II. RELATED WORK

A. Holistic Co-Speech Motion Generation

Early co-speech gesture generation mainly learned direct mappings from speech to body motion, considering person-specific behavior, multimodal cues, speaker identity, and gesture style [16]–[22]. Subsequent studies addressed the one-to-many nature of speech-to-motion mapping through probabilistic latent variables and learned gesture prototypes [13], [14], while improving cross-modal association, zero-shot style control, diffusion-based synthesis, and quantization-based motion matching [23]–[27].

With large-scale datasets such as BEAT [28], recent research has shifted toward holistic generation involving face, body, hands, and global motion. Representative methods employ structured discrete motion representations [1], [2], [29], diffusion or efficient sequence modeling [3], [30], and explicit cross-region or multi-scale dependencies [4], [5]. GlobalDiff [31] reduces hierarchical error accumulation through global-rotation modeling, while HIP [32] and LoM [33] further explore hierarchical periodicity and unified verbal–non-verbal motion modeling. Recent systems also extend generation toward causal, streaming, and interactive settings [34], [35]. In contrast to primarily improving the low level generator, IntentTalk introduces an intermediate representation of *upcoming body-motion organization* between speech understanding and detailed motion realization.

B. Semantic and Intention-Aware Gesture Generation

Semantic compatibility has long been recognized as complementary to motion realism. Early multimodal methods incorporated linguistic cues together with speech acoustics [18], [20], followed by approaches that explicitly separate or refine semantic and rhythmic information [36], [37]. Semantic guidance has also been introduced through cross-modal latent spaces and multimodal diffusion [38], [39].

Another line of research addresses sparse semantic gestures using retrieval or external priors. Semantic Gesticulator, RAG-Gesture, and GestureHYDRA retrieve or inject explicit semantic knowledge [6], [11], [40], while ExGes [41] employs retrieved fine-grained motion references. Without relying primarily on retrieval, SemGes [8] models multi-granularity semantics and SemTalk [7] selectively emphasizes sparse semantically relevant motion.

More recent studies introduce higher-level semantic organization through communicative intentions [10], coordinated semantic and beat streams [9], gesture captions and motion-language modeling [42], and contrastive holistic semantic grounding [43]. Related approaches further exploit hierarchical motion representations, language-level abstractions of gesture form and function, or trajectory-level action control [44]–[46]. IntentTalk differs in the construction of its intermediate variable: instead of using predefined semantic categories, retrieved exemplars, captions, or explicit intention descriptions, it derives discrete planning targets from an offset window of upcoming body motion and learns to predict them from speech. The resulting representation is therefore *motion-grounded* and *future-oriented*, without requiring future motion at inference.

C. Discrete Motion Representation and Future Planning

Discrete representations provide compact abstractions of continuous signals [47] and have been widely adopted for human-motion generation. In co-speech gesture synthesis, learned prototypes and vector-quantized representations support motion variability, matching, compositional body–hand modeling, and holistic motion decomposition [1], [14], [27], [29]. ProbTalk [2] further models coordinated variability

through product quantization, while MambaTalk [3] combines discrete priors with efficient sequence modeling.

Discrete tokenization is also widely used in text-driven motion generation, including autoregressive motion-token prediction [48], [49], masked generative modeling [50], [51], and hierarchical whole-body motion representation [52]; diffusion-based models provide another strong generative paradigm for text- and speech-driven motion [25], [53]. Unlike prior tokenizers that mainly represent motion for reconstruction, matching, or generation, IntentTalk deliberately excludes the current motion token and encodes an offset window of subsequent body motion into a discrete target. The resulting token summarizes upcoming motion dynamics, and a speech-side predictor learns to infer it before detailed generation, thereby forming an explicit future-planning stage between speech conditioning and motion realization.

III. METHOD

A. Overview

Given multimodal speech observations S , speaker identity I , and a short seed motion M_{seed} , our goal is to generate a holistic motion sequence $M = \{m_n\}_{n=1}^N$ containing coordinated facial, upper-body, hand, and lower-body motions. Existing approaches commonly model the direct mapping $p(M | S, I, M_{\text{seed}})$, in which semantic organization and low-level motion realization are handled within a single generation process. Such a formulation does not explicitly represent the near-future organization of a semantic gesture before its detailed realization.

IntentTalk introduces a sequence of discrete future motion plans $P = \{k_t\}_{t=1}^T$ as an intermediate representation between speech and motion. The generation process is formulated as

$$p(M | S, I, M_{\text{seed}}) = \sum_P p(P | S) p(M | P, S, I, M_{\text{seed}}), \quad (1)$$

where $p(P | S)$ predicts motion-grounded plans from speech, while $p(M | P, S, I, M_{\text{seed}})$ realizes detailed motion conditioned on the predicted plans. We use the term *motion plan* to denote a latent representation of upcoming body-motion organization rather than an estimate of psychological intention.

As illustrated in Fig. 2, IntentTalk consists of three main components. First, a future motion-plan tokenizer learns a discrete plan vocabulary from ground-truth future motion. Second, a speech-to-plan predictor infers the corresponding plan tokens from speech-semantic features. Finally, the predicted plan guides motion generation through two complementary mechanisms: an event-aware semantic gate regulates *when* semantic motion is activated, while hierarchical FiLM modulation controls *how* the planned motion is realized. Future motion is used only to construct training targets and is not required during inference.

B. Future-Grounded Discrete Motion Planning

1) *Future motion-plan tokenizer*: The holistic motion backbone models facial and body motion through separate generation pathways. Since the proposed planning mechanism is designed to capture the temporal organization of communicative

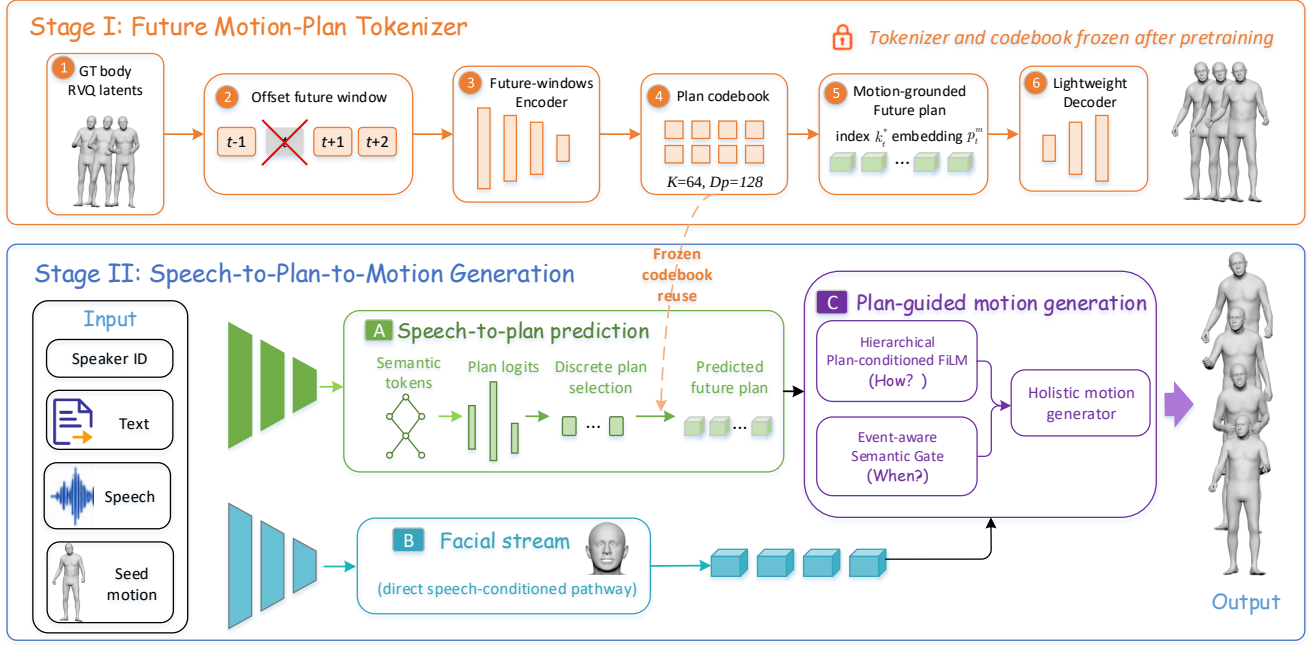


Fig. 2. Overview of IntentTalk. Stage I learns a motion-grounded discrete plan vocabulary from offset future-motion windows. Stage II predicts future plans from speech and uses them to regulate semantic activation and hierarchical body-motion realization.

body gestures, we construct the motion plan from the upper-body, hand, and lower-body residual vectorquantized autoencoder (RVQ-VAE) latents, while the facial stream retains the original speech-aligned generation pathway. Let $z_t^{r,q} \in \mathbb{R}^{D_m}$ denote the latent of body region r at motion-token position t and RVQ-VAE level q .

$$\begin{aligned} z_t^r &= \sum_q z_t^{r,q}, \quad r \in \{\text{upper, hands, lower}\}, \\ u_t &= [z_t^{\text{upper}}, z_t^{\text{hands}}, z_t^{\text{lower}}], \\ W_t^+ &= \text{Vec}(u_{t+o}, \dots, u_{t+o+H-1}). \end{aligned} \quad (2)$$

Each regional latent has 256 dimensions, yielding a 768-dimensional body token. We use a future horizon of $H = 2$ and an offset of $o = 1$, such that the current motion token is excluded from the plan target. Since each RVQ-VAE token corresponds to four frames at 30 FPS, W_t^+ covers the subsequent eight frames (≈ 0.267 s). Positions without a complete future window are excluded from plan-related objectives.

A future-window encoder E_m first maps each future motion window W_t^+ to a continuous representation, $h_t^m = E_m(W_t^+)$. We then discretize h_t^m using a learnable codebook $\mathcal{C} = \{c_k\}_{k=1}^K$ by nearest-neighbor quantization:

$$k_t^* = \arg \min_k \|h_t^m - c_k\|_2^2, \quad p_t^m = c_{k_t^*}, \quad (3)$$

where k_t^* denotes the discrete plan index, and p_t^m is the corresponding motion-grounded plan embedding. We use $K = 64$ plan codes with embedding dimension $D_p = 128$. A lightweight decoder reconstructs the complete future window from p_t^m . The tokenizer is optimized as

$$\mathcal{L}_{\text{VQ}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{code}} + \beta_{\text{com}} \mathcal{L}_{\text{com}}, \quad (4)$$

where $\mathcal{L}_{\text{rec}}$ denotes masked future-window reconstruction, and $\mathcal{L}_{\text{code}}$ and $\mathcal{L}_{\text{com}}$ are the standard codebook and commitment

losses. After pretraining, the tokenizer and codebook are frozen and provide motion-derived plan targets k_t^* during subsequent training.

2) *Speech-to-plan prediction*: Future motion is unavailable at inference. We therefore learn a speech-side planner to predict the motion-derived plan tokens using only speech information. The semantic branch integrates linguistic features, utterance-level CLIP text features, and temporally aligned HuBERT representations to obtain frame-level semantic features $\{f_n^{\text{sem}}\}_{n=1}^N$. Every four consecutive features are averaged into a semantic token s_t , matching the temporal resolution of the RVQ motion tokens.

The semantic-token sequence is processed by a two-layer Transformer encoder E_s to obtain contextual representations $h_{1:T}^s = E_s(s_{1:T})$. A linear prediction head then produces K -way plan logits $l_t = W_p h_t^s + b_p$. Rather than averaging multiple codebook entries, we select a single discrete plan at each temporal position. Specifically, the selected plan embedding is

$$\begin{aligned} \hat{p}_t &= a_t^\top C, \\ a_t &= \begin{cases} \text{STGumbelSoftmax}(l_t, \tau_p), & \text{training,} \\ \text{OneHot}(\arg \max_k l_{t,k}), & \text{inference,} \end{cases} \end{aligned} \quad (5)$$

where $C \in \mathbb{R}^{K \times D_p}$ is the frozen motion-plan codebook. During training, straight-through Gumbel-Softmax provides a discrete one-hot selection in the forward pass while preserving gradient propagation. At inference, the plan with the largest logit is selected deterministically.

The speech planner is supervised by the motion-derived plan index k_t^* . We employ a semantic-weighted classification loss $\mathcal{L}_{\text{cls}}$ to encourage correct plan prediction, assigning larger weights to temporally valid positions with higher semantic relevance. In addition, a cosine alignment loss $\mathcal{L}_{\text{align}}$ encourages the predicted embedding $\hat{p}_t$ to approach its target code $c_{k_t^*}$.

A fixed-codebook contrastive objective $\mathcal{L}_{\text{ctr}}$ further aligns the continuous speech representation h_t^s with the corresponding motion-plan code while distinguishing it from the remaining codebook entries. Positions without a valid future-motion target are excluded from all plan-related objectives.

C. Plan-Guided Motion Generation

The predicted plan $\hat{p}_t$ affects motion synthesis through two complementary pathways. The event-aware semantic gate regulates the temporal activation of semantic motion, whereas hierarchical FiLM modulates its detailed body realization.

1) *Event-aware semantic gating*: To regulate the contribution of semantic information during motion generation, we first estimate a two-class semantic gate from the speech representation: $g_t^s = G_{\text{sem}}(s_t) \in \mathbb{R}^2$, where the two logits correspond to inactive and active semantic-motion states. This speech-derived gate estimates semantic relevance at the current temporal position, but does not explicitly incorporate the predicted organization of upcoming motion. We therefore introduce an event-aware refinement conditioned on the future motion plan.

Specifically, an event predictor jointly encodes the semantic representation s_t and the predicted plan $\hat{p}_t$:

$$e_t = E_{\text{event}}([s_t; \hat{p}_t]). \quad (6)$$

The predicted plan further produces a bounded gate residual

$$\Delta g_t^p = \tanh(W_\delta \hat{p}_t + b_\delta), \quad (7)$$

while the event prediction provides an activation-oriented correction

$$\Delta g_t^e = \frac{1}{2} \tanh(e_t)[-1, 1]. \quad (8)$$

The refined semantic gate is then computed as

$$\tilde{g}_t = g_t^s + \alpha_g (\Delta g_t^p + \Delta g_t^e). \quad (9)$$

The refined gate is converted into an activation probability as $\pi_t^{\text{act}} = \text{Softmax}(\tilde{g}_t)_{\text{active}}$. The activation probability is repeated over the corresponding four frames and multiplicatively applied to the semantic body features before cross-modal decoding. In this way, the speech-derived semantic estimate provides the primary activation cue, while the predicted future plan introduces a bounded temporal refinement.

The event predictor is supervised by a binary cross-entropy loss $\mathcal{L}_{\text{event}}$. The event target is obtained from the frame-level semantic annotations by averaging the annotations within each four-frame interval and assigning a positive label when the resulting value is greater than zero. The output layers of the event predictor and the plan-conditioned gate residual are initialized to zero. Consequently, $\tilde{g}_t = g_t^s$ at initialization, allowing the plan-aware refinement to be introduced gradually during optimization.

2) *Hierarchical plan-conditioned FiLM*: While the event-aware gate regulates *when* semantic motion is activated, the predicted plan further controls *how* the body motion is realized. For a feature X_t^l at generation level l , a level-specific projection maps the predicted plan $\hat{p}_t$ to FiLM parameters:

$$[\gamma_t^l; \beta_t^l] = F_l(\hat{p}_t), \quad (10)$$

where γ_t^l and β_t^l denote the scale and shift terms, respectively. The plan-conditioned feature is then computed as

$$\tilde{X}_t^l = X_t^l \odot [1 + \tanh(\gamma_t^l)] + \beta_t^l. \quad (11)$$

When the temporal resolutions of the plan and motion features differ, the plan sequence is linearly interpolated to match the corresponding feature rate. The residual scaling term keeps the modulation bounded, while the FiLM projections are zero-initialized so that $\tilde{X}_t^l = X_t^l$ at initialization.

Plan conditioning is introduced at three levels of body-motion generation. At the *shared body level*, the common temporal representation is modulated before anatomical decomposition. At the *body-part level*, independent FiLM modules condition the upper-body, hand, and lower-body features before cross-part interaction. At the *motion-token level*, the regional RVQ latents are further modulated before residual code prediction. The plan is applied only to body-motion branches, while facial motion remains directly conditioned on speech without plan modulation.

D. Training and Inference

1) *Training*: IntentTalk is trained in two stages. In Stage I, the future motion-plan tokenizer is optimized on ground-truth RVQ latents using $\mathcal{L}_{\text{VQ}}$. After convergence, the tokenizer and its codebook are frozen. In Stage II, the frozen tokenizer provides motion-derived plan targets k_t^* and validity masks, while the speech-to-plan predictor, event-aware gating module, hierarchical FiLM modules, and motion generator are jointly optimized.

Let $\mathcal{L}_{\text{gen}}$ denote the motion-generation objective, including body-latent reconstruction, residual RVQ prediction, and semantic-motion learning. The plan-related objective is defined as

$$\mathcal{L}_{\text{plan}} = \mathcal{L}_{\text{cls}} + \lambda_a \mathcal{L}_{\text{align}} + \lambda_c \mathcal{L}_{\text{ctr}} + \lambda_e \mathcal{L}_{\text{event}}. \quad (12)$$

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{gen}} + \eta_j \lambda_p \mathcal{L}_{\text{plan}}, \quad (13)$$

where $\eta_j \in [0, 1]$ denotes an epoch-dependent schedule that gradually introduces plan-related supervision after an initial warm-up period. Detailed loss weights and scheduling parameters are provided in the experimental setup.

2) *Inference*: At inference, the future-window encoder and decoder are removed, while the learned plan codebook is retained. Given speech, speaker identity, and a short seed motion, the speech-to-plan predictor estimates a discrete plan token at each plan temporal position. The corresponding codebook embedding $\hat{p}_t$ refines semantic activation through the event-aware gate and conditions body-motion realization through hierarchical FiLM. The motion generator then predicts the regional RVQ codes and reconstructs the final holistic motion sequence. Therefore, ground-truth future motion is used only to construct motion-plan supervision during training and is not required at inference.

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON BEAT2 UNDER THE ONE-SPEAKER AND ALL-SPEAKER SETTINGS.

Method	One-Speaker						All-Speaker					
	FGD↓	BC↑	DIV↑	SRGR↑	MSE↓	LVD↓	FGD↓	BC↑	DIV↑	SRGR↑	MSE↓	LVD↓
EMAGE [1]	5.512	0.772	13.06	0.323	7.680	7.556	5.643	<u>0.771</u>	<u>12.92</u>	0.293	6.908	7.593
ProbTalk [2]	5.040	0.771	13.27	0.305	8.617	7.634	5.827	0.744	11.82	0.283	6.621	6.965
MambaTalk [3]	5.366	0.781	13.05	0.297	<u>6.289</u>	6.897	5.316	0.746	11.64	0.276	7.875	7.792
RAG-Gesture [6]	8.082	0.734	11.97	0.395	7.248	6.947	4.871	0.647	11.42	0.371	7.262	6.838
SemTalk [7]	4.278	0.777	12.91	0.434	6.153	6.938	3.562	0.769	12.74	0.409	7.612	7.498
SemGes [8]	4.441	0.685	12.17	0.428	6.754	7.018	3.382	0.649	11.17	0.404	6.786	<u>6.673</u>
GestureLSM [4]	4.247	0.729	13.76	0.413	7.021	7.102	3.268	0.725	11.20	0.385	6.963	7.081
PyraMotion [5]	4.612	0.742	13.42	0.411	7.176	7.270	3.581	0.751	12.74	0.398	6.857	7.155
GlobalDiff [31]	4.780	0.705	13.73	<u>0.435</u>	6.330	6.787	2.630	0.740	12.24	0.407	4.144	7.252
DuoGesture [9]	4.101	0.756	12.34	0.422	7.103	7.646	4.081	0.770	12.83	<u>0.414</u>	6.802	7.658
MIBURI [35]	7.538	0.791	13.67	0.397	8.361	7.895	3.805	0.748	12.79	0.388	7.772	7.543
LiveGesture [34]	4.574	<u>0.794</u>	<u>13.91</u>	0.412	6.884	6.917	3.378	0.732	12.54	0.381	7.648	7.247
StreamTalk [12]	<u>3.834</u>	0.704	13.18	0.416	6.874	<u>6.677</u>	2.934	0.616	12.47	0.408	6.845	6.741
IntentTalk (Ours)	3.812	0.817	13.95	0.442	6.608	6.462	<u>2.661</u>	0.799	13.25	0.425	<u>6.524</u>	6.471

IV. EXPERIMENTS

A. Experimental Settings

1) *Dataset*: We train and evaluate our model on BEAT2 [1], using its English subset and following the official 85:7.5:7.5 split protocol for training, validation, and testing. To assess cross-dataset generalization, we further perform zero-shot evaluation on GES-X [15], a large-scale in-the-wild co-speech gesture dataset containing diverse speakers and recording conditions. The GES-X data are used only for evaluation and are not involved in model training or fine-tuning. This setting allows us to examine the robustness of the learned motion-planning and generation mechanism under distribution shifts in speakers, motion patterns, and capture conditions.

2) *Implementation Details*: All experiments are conducted on a single NVIDIA A100 GPU. Following EMAGE [1], motions are sampled at 30 FPS using the same SMPL-X representation, and audio is sampled at 16 kHz. HuBERT, fastText, and CLIP features are used for speech-semantic modeling. Training clips contain 64 motion frames with four seed frames. The body backbone has a hidden dimension of 768, while the speech planner is a two-layer Transformer with 128-dimensional features. The future-plan tokenizer contains two residual MLP blocks and a 64-entry, 128-dimensional codebook, with $H = 2$ and $o = 1$. We pretrain the tokenizer for 100 epochs with a learning rate of 2×10^{-4} and then freeze it while training the full model for 200 epochs at 5×10^{-5} . Plans are selected using straight-through Gumbel-Softmax during training and argmax at inference. The remaining optimization hyperparameters and loss weights are given in the supplementary material.

3) *Evaluation Metrics*: We comprehensively evaluate the generated motions in terms of motion realism, diversity, speech-motion synchronization, semantic relevance, and facial motion accuracy. Fréchet Gesture Distance (FGD) [18] measures the distribution discrepancy between generated and ground-truth body gestures in a learned feature space, with a lower value indicating better motion realism. Gesture diversity is evaluated using Diversity (DIV) [13], computed from the

average feature-space distance among generated motion clips. Beat Consistency (BC) [54] assesses the temporal alignment between speech beats and motion beats, while Semantic-Relevant Gesture Recall (SRGR) [28] evaluates whether generated gestures reproduce semantically relevant motion patterns emphasized by the corresponding semantic annotations. For facial motion, we adopt vertex Mean Squared Error (MSE) [26] and Lip Vertex Distance (LVD) [29] to quantify positional reconstruction error and discrepancies in lip-region motion between the generated and ground-truth facial vertices, respectively.

4) *Baselines*: We compare IntentTalk with thirteen representative co-speech motion generation methods spanning complementary modeling paradigms. These include quantization-based holistic approaches EMAGE [1] and ProbTalk [2]; efficient temporal models MambaTalk [3], GestureLSM [4], and PyraMotion [5]; semantic-aware methods RAG-Gesture [6], SemTalk [7], SemGes [8], and DuoGesture [9]; and GlobalDiff [31], which models motion through global-rotation diffusion. We further include recent online and streaming methods MIBURI [35], LiveGesture [34], and StreamTalk [12]. StreamTalk is particularly relevant because it employs future key-pose anchoring, providing a direct reference for evaluating the distinction between future anchoring and the proposed future-window planning.

B. Quantitative Analysis

1) *Comparison with State-of-the-Art Methods*: Table I compares IntentTalk with representative methods on BEAT2 under both one-speaker and all-speaker settings. In the one-speaker setting, IntentTalk achieves the best FGD (3.812), BC (0.817), DIV (13.95), SRGR (0.442), and LVD (6.462), outperforming the strongest competing methods on most metrics. The simultaneous improvements in BC and SRGR indicate that future-grounded planning enhances both rhythmic synchronization and semantic relevance. Although its MSE (6.608) is not the lowest, this suggests that the performance gain is not simply due to improved frame-wise reconstruction.

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE GES-X
DATASET.

Method	FGD↓	BC↑	DIV↑	SRGR↑	MSE↓	LVD↓
EMAGE [1]	27.512	0.612	10.86	0.203	38.680	32.556
ProbTalk [2]	24.840	0.631	11.27	0.185	40.117	33.634
MambaTalk [3]	26.366	0.651	11.05	0.197	32.289	29.897
RAG-Gesture [6]	32.082	0.604	9.97	0.215	37.248	30.947
SemTalk [7]	21.278	0.647	10.91	0.244	<u>31.153</u>	28.938
SemGes [8]	22.441	0.585	10.17	0.228	33.754	29.018
GestureLSM [4]	20.847	0.619	11.76	0.253	35.021	29.102
PyraMotion [5]	22.612	0.622	11.42	0.241	34.176	30.270
GlobalDiff [31]	23.780	0.595	11.73	0.265	32.330	28.787
DuoGesture [9]	19.501	0.636	10.34	0.252	34.103	31.646
MIBURI [35]	29.538	0.661	11.67	0.227	39.361	32.895
LiveGesture [34]	21.574	0.674	<u>11.91</u>	0.232	32.884	28.917
StreamTalk [12]	<u>18.367</u>	<u>0.705</u>	11.88	<u>0.288</u>	32.470	29.040
IntentTalk (Ours)	17.812	0.717	12.15	0.292	29.608	26.462

Under the more challenging all-speaker setting, IntentTalk achieves the best BC (0.799), DIV (13.25), SRGR (0.425), and LVD (6.471), while obtaining the second-best FGD of 2.661, close to GlobalDiff (2.630). These results show that the proposed planning representation generalizes across speakers and provides effective guidance for semantic and temporal motion organization. Overall, IntentTalk exhibits a more balanced advantage in motion quality, synchronization, diversity, and semantic consistency than methods optimized primarily for reconstruction-oriented metrics.

2) *Cross-Dataset Generalization*: Table II reports cross-dataset results on GES-X. Without additional adaptation, IntentTalk achieves the best results across all six metrics, including an FGD of 17.812, BC of 0.717, DIV of 12.15, and SRGR of 0.292. Compared with the strongest baselines, it further reduces MSE to 29.608 and LVD to 26.462.

The consistent improvements under cross-dataset distribution shifts indicate that the learned future-plan representation captures transferable motion structure rather than dataset-specific speech–motion correlations. The gains over both conventional speech-to-motion methods and recent future-aware approaches further support the effectiveness of segment-level future planning for robust co-speech gesture generation.

3) *Ablation Study*: Table III investigates the contribution of the proposed planning components. Replacing the future-grounded plan with the current motion token consistently degrades all metrics, increasing FGD from 3.812 to 4.286 and reducing SRGR and Event F1 from 0.442 and 0.784 to 0.429 and 0.706, respectively. This result indicates that representing upcoming motion dynamics provides more informative guidance than simply encoding the current motion state.

Removing the event planner leads to a clear drop in Event F1 (0.784→0.687) and SRGR (0.442→0.433), confirming its importance for determining the temporal activation of semantically relevant motion. In contrast, removing plan modulation mainly affects the realization quality of the planned gesture: FGD increases to 4.181 and DIV decreases to 13.67, while Event F1 remains relatively high at 0.762. This complementary

TABLE III
ABLATION STUDY OF INTENTTALK.

Variant	FGD↓	BC↑	DIV↑	SRGR↑	Event F1↑
Current Motion Token	4.286	0.798	13.61	0.429	0.706
Future Plan Token	4.137	0.807	13.78	<u>0.436</u>	0.748
w/o Event Planner	<u>4.126</u>	0.809	<u>13.80</u>	0.433	0.687
w/o Plan Modulation	4.181	<u>0.812</u>	13.67	0.435	<u>0.762</u>
Full IntentTalk	3.812	0.817	13.95	0.442	0.784

TABLE IV
EFFECT OF DIFFERENT PLANNING CONDITIONS AND FUTURE GUIDANCE
REPRESENTATIONS ON INTENTTALK.

Plan Condition	FGD↓	SRGR↑	Event F1↑	Acc@1↑	Acc@5↑
No Plan	4.347	0.424	0.668	–	–
Random Plan	4.412	0.418	0.641	–	–
Current Motion Token	4.286	0.429	0.706	0.462	0.731
Future Key-Pose Anchor	4.156	0.435	0.742	<u>0.498</u>	<u>0.764</u>
Predicted Future Plan	<u>4.012</u>	<u>0.442</u>	<u>0.784</u>	0.536	0.798
Oracle Future Plan	3.846	0.457	0.836	–	–

behavior is consistent with the design of the two modules, where the event planner primarily determines *when* semantic motion should be activated and hierarchical plan modulation controls *how* the planned motion is realized. The full model achieves the best performance across all metrics, demonstrating that future-grounded planning, event-aware activation, and hierarchical modulation provide complementary benefits when jointly integrated. Additional qualitative ablation comparisons are provided in the supplementary material.

4) *Effect of Different Planning Conditions and Future Guidance*: Table IV examines whether the gains of IntentTalk originate from meaningful future-oriented planning rather than merely adding an auxiliary latent condition. Removing the plan degrades FGD, SRGR, and Event F1 to 4.347, 0.424, and 0.668, respectively, while a random plan further reduces performance to 4.412, 0.418, and 0.641. This indicates that the improvement depends on motion-relevant planning information rather than the presence of an additional conditioning vector.

We further compare current-state conditioning, future anchoring, and future-window planning under the same backbone and training protocol. The Current Motion Token obtains 4.286 FGD, 0.429 SRGR, and 0.706 Event F1, which improve to 4.156, 0.435, and 0.742 with the Future Key-Pose Anchor and further to 4.012, 0.442, and 0.784 with the Predicted Future Plan. This progressive trend suggests that a future endpoint provides useful forward guidance, while the window-level plan offers additional information about the temporal evolution of upcoming motion. Acc@1/Acc@5 similarly increase from 0.462/0.731 to 0.498/0.764 and 0.536/0.798, respectively, providing complementary evidence that the future-oriented discrete targets are predictable from speech.

Finally, the Oracle Future Plan achieves the best FGD (3.846), SRGR (0.457), and Event F1 (0.836), providing an empirical upper bound with accurate future planning targets. The remaining gap to the predicted setting suggests further

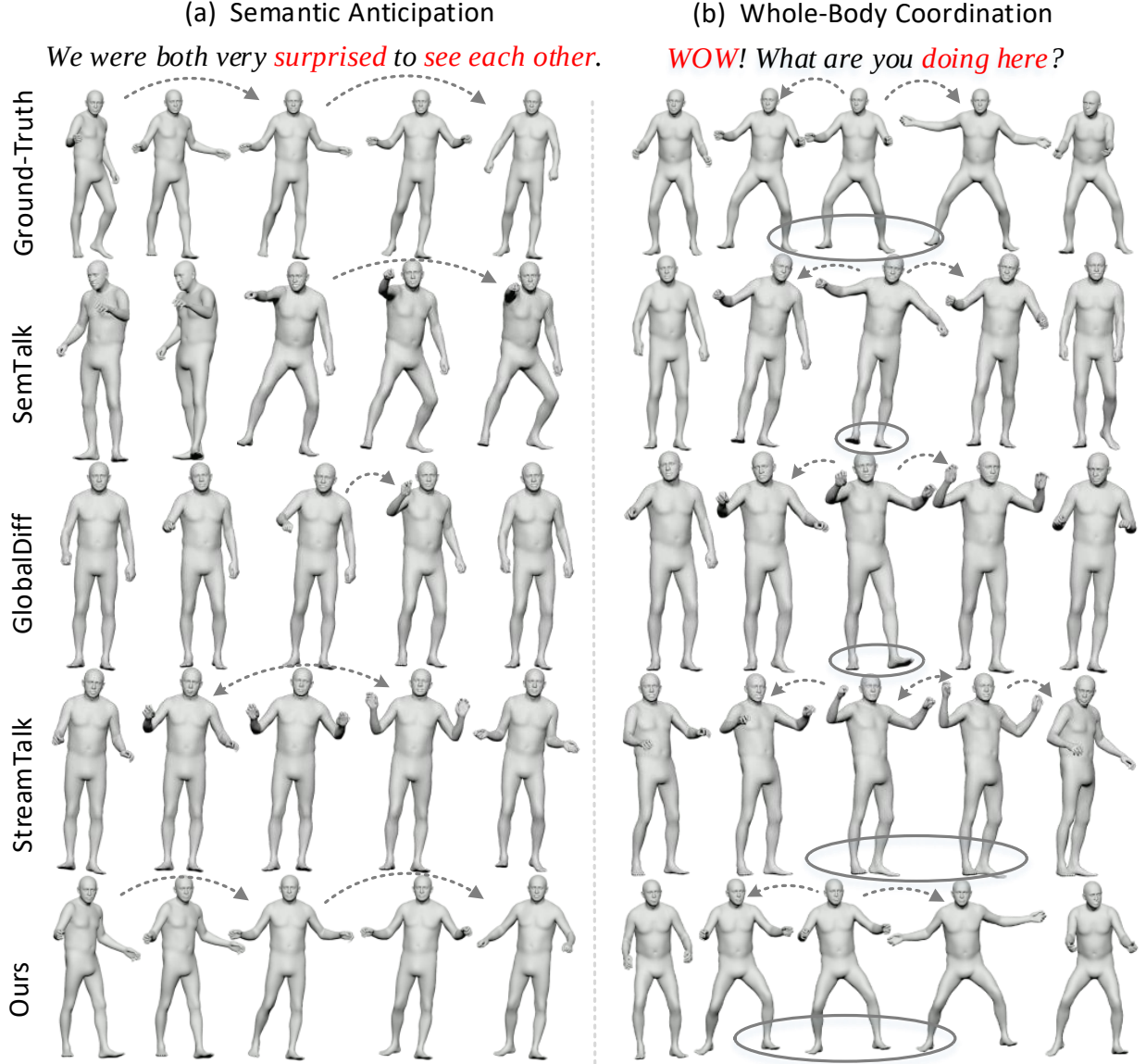


Fig. 3. Qualitative comparison of representative co-speech motion sequences. (a) Semantic anticipation around a salient speech event, where IntentTalk exhibits earlier preparation and more coherent gesture evolution. (b) Whole-body coordination, where IntentTalk produces more coordinated upper-body, hand, torso, and lower-body motion. All methods are compared around the same speech events.

room for improving speech-to-plan prediction. Overall, these results support a progressive benefit from current-state conditioning to future anchoring and further to temporally extended future planning.

C. Qualitative Analysis

1) *Qualitative Comparison with State-of-the-Art Methods:* Fig. 3 compares representative motion sequences in terms of semantic anticipation and whole-body coordination. As shown in Fig. 3(a), the reference motion develops gradually from preparatory arm movement toward a salient bilateral gesture. SemTalk tends to produce a more localized response, GlobalDiff exhibits relatively limited semantic-specific variation, and StreamTalk generates smooth but more generic bilateral motion. In comparison, IntentTalk presents a clearer progression from preparation to semantic realization, suggesting

that the predicted future plan provides useful guidance for organizing motion before detailed generation. In Fig. 3(b), the reference gesture involves coordinated changes in the arms, torso, and lower-body stance. Competing methods mainly capture the upper-body response, whereas IntentTalk exhibits more consistent cross-part evolution, with bilateral arm expansion accompanied by a corresponding stance change. These qualitative observations support the role of future planning in improving both temporal gesture organization and holistic body-part coordination.

D. User Study

To complement the objective evaluation, we conduct a perceptual user study comparing IntentTalk with SemTalk, GlobalDiff, and StreamTalk. We recruit 30 participants to

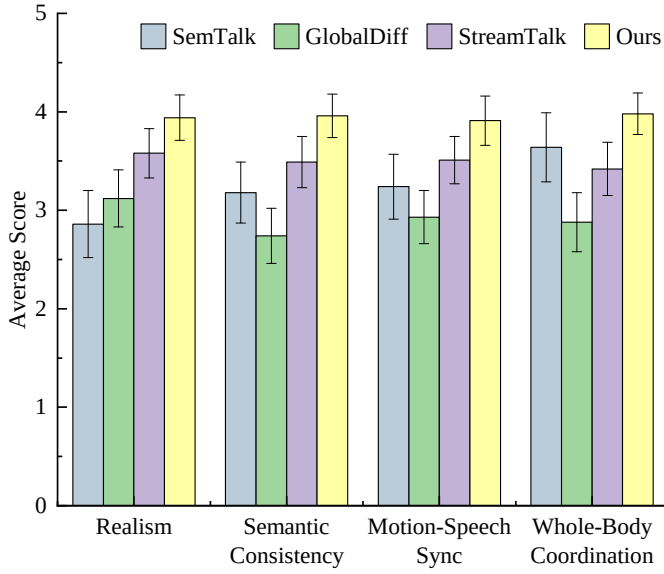


Fig. 4. User study results based on a five-point MOS scale.

provide Mean Opinion Scores (MOS) on a five-point Likert scale across realism, semantic consistency, motion–speech synchronization, and whole-body coordination. Ten test sequences are randomly selected for each method, resulting in 40 videos and 300 evaluations per method. Inter-rater reliability is measured using Krippendorff’s alpha ($\alpha = 0.78$), and pairwise significance is assessed using Wilcoxon signed-rank tests ($p < 0.01$). Additional protocol details are provided in the supplementary material.

As shown in Fig. 4, IntentTalk obtains the highest MOS across all four criteria, scoring 3.94, 3.96, 3.91, and 3.98, respectively. Compared with StreamTalk, the strongest baseline, the corresponding gains are 0.36, 0.47, 0.40, and 0.56 points. The improvements in semantic consistency and whole-body coordination support the effectiveness of future planning for organizing upcoming motion, while the gain in motion–speech synchronization is consistent with event-aware temporal activation. These subjective results further corroborate the perceptual benefits of the proposed framework.

V. CONCLUSION

In this paper, we proposed IntentTalk, a future-grounded speech-to-plan-to-motion framework that explicitly models upcoming gesture organization before detailed motion realization. A discrete motion-plan space is learned from future motion windows and predicted from speech alone at inference, while event-aware semantic gating and hierarchical plan-conditioned modulation respectively regulate when semantic motion is activated and how the predicted plan is realized across body regions. Experiments on BEAT2 and zero-shot evaluation on GES-X demonstrate consistent improvements in semantic relevance, temporal organization, and holistic motion coordination, together with competitive motion quality, synchronization, and diversity. Further planning and ablation analyses show that future-oriented plans provide more informative guidance than current-state or unstructured representa-

tions, while the remaining gap to oracle plans indicates room for improved speech-to-plan prediction. These results support explicit future motion planning as an effective intermediate structure for co-speech gesture generation. Future work will explore longer-horizon and probabilistic planning, as well as causal planning for real-time interactive generation.

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